The Development of Soft Systems Methodology through the Lens of the FMA Framework

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Abstract

The 30-year Action Research programme conducted at the University of Lancaster, by Peter Checkland and colleagues, stands as the most rigorous ever undertaken in the field of Systems Thinking. It began with Systems Engineering as the chosen methodology and culminated with the establishment of the mature form of Soft Systems Methodology. This paper reviews the history of that programme through the lens of the FMA framework proposed by Checkland and Holwell as a benchmark for effective AR. In their view, the cycle of AR must include a declaration of the *framework of ideas* (F) and the *methodology* (M) used to intervene in an *area of concern* (A). It should yield both action in the A and findings about the F and M. The paper applauds the success of the programme in terms of the interventions undertaken in complex As and the modifications made to the M because of the learning gained from those interventions. It notes, however, that there were no further significant revisions to the F once the break from Systems Engineering was achieved. This was beneficial because it enabled Soft Systems Methodology to continue to refine its unique competence in a new type of A, thereby significantly expanding the repertoire of Systems Thinking. There were consequences, however, because it imposed constraints on the usefulness of the M in other types of A. On this basis, it is suggested that Systems Thinking as a field will only reach its full potential when it is prepared to entertain a variety of Fs and Ms, ensuring its relevance to the broadest possible range of As.

Keywords: Soft Systems Methodology, Action Research, FMA framework

Introduction

This paper charts the development of Soft Systems Methodology (SSM) using the FMA framework developed by Checkland and Holwell (1998a, 1998b) as a benchmark for carrying out Action Research (AR). It begins by outlining Checkland's commitment to AR and describing the nature of the FMA framework. The framework is then used to trace the development of SSM through five transitions – *'Systems Engineering (SE) to Blocks and Arrows', 'Blocks and Arrows to Seven-Stages', 'Seven-Stages to Two Streams', 'Two Streams to Four Main Activities', and 'Four Main Activities and Beyond'*. The landmarks along this route are marked by SE and the four representations of SSM identified by Holwell (1997) as in use between 1972 and 1990. These were confirmed by Checkland as appropriate in his 30-year retrospective on SSM (Checkland 1999). The FMA model draws attention to the *framework of ideas* underpinning a methodology, the *methodology* itself, and the *area of application* in which it is used. Clarifying these elements for SE and each representation of SSM allows changes that take place in between the landmarks to be detected and reasons for the changes described based on interactions that occurred between the elements. A review follows of how successful SSM has been in extending its As and developing its M and F because of employing the FMA framework. Particular attention is given to SSM's F because it has been under-researched and its crystallization, after the break with Systems Engineering, has much to tell us about how AR should in future be conducted in Systems Thinking. The lessons are drawn out in a conclusions section which confirms the value of the FMA framework for delivering recoverable results through AR, considers the advantages and disadvantages of SSM's employment of a settled F, and draws out the implications for AR as a vehicle for advancing Systems Thinking as a field of research and practice.

The paper is the first to examine the development of SSM through the lens of the FMA framework. In particular, it is original in giving detailed consideration to the development, or lack of development, of the F underpinning SSM and how this has impacted on the M and the

A. There is a gap in the literature here. Houghton and Ledington (2002) bemoan the lack of attention that has been given to the social reality implied by SSM, find confusion in the literature that does exist, and argue that this is something ripe for further research efforts. Stowell (2024) regrets the lack of discussion of SSM's F and blames the very success of the M for this situation. He has, as we shall see, sought to revitalise debate on this issue. Other excellent accounts of the development of SSM are those of Holwell (1997), favourably referenced by Checkland (1999), and Mingers (2000). Use of the FMA framework allows their accounts to be supplemented, particularly regarding the issues that arise from the lack of attention given to developing the F underlying SSM.

A paper of this type requires the author to explain the F they are themselves employing in conducting the analysis. For the most part, I have tried to inhabit the F set out by Checkland at the various stages of the development of SSM. The account is based on a sympathetic reading of his books and papers. It is there to be verified, corrected or supplemented by other scholars. In the 'Review' section, however, I bring alternative Fs to bear with a view to revealing, from the functionalist and radical change perspectives, the constraints imposed by Checkland's F on the further development of his M and on its competence in meeting the challenges posed by the range of possible As. My own F therefore, justified I feel by the results it yields in this paper, is that it is useful to view any methodology and what it can achieve from a variety of different perspectives.

Action Research (AR) and the FMA Framework

Professor Gwilym Jenkins, who established the Department of Systems Engineering at Lancaster University in the UK in the mid-1960s, stated in his inaugural lecture that:

> Like the Department of Operational Research and Marketing, the Systems Engineering Department is involved in outside projects which aim to provide experience in the application of systems engineering for staff and students alike. At the moment we are

> concerned with nine such projects drawn from the oil, chemical, steel, shipbuilding and electrical industries (Jenkins 1967, p.10).

Thus, when Checkland joined the department in 1969 he was entering an environment in which taking ideas out into the real-world to test them was already the norm. He asserted his corresponding commitment to engaging in practical projects in his own inaugural lecture:

> The Systems Engineer is himself ready, willing, and anxious, to implement the proposed system, to appraise it after a suitable operational period and to re-design if necessary … systems studies are carried out in collaboration with managers, and in continuing dialogue with them (Checkland 1969, p. 15).

Firm foundations for an AR programme were put in place with the launch of a one-year Masters course in 'Systems in Management', which included a five-month project, and, later, the creation of a university-owned consultancy company, called ISCOL, which could take on projects at all times of the year and of any duration. As Checkland remarked:

> It seemed to me … that the arrangement we had at Lancaster provided a remarkable opportunity to mount a research *programme* comprising many different *projects* in organizations of many different kinds, in both the private and public sectors. Thus began the development of what emerged as Soft Systems Methodology (SSM) … (Checkland 2011, p. 497; italics in the original).

In 1972, Checkland clearly identified the research on which he was engaged as an AR programme, started by Jenkins in 1966, in which 'more than a hundred such projects have now been carried out' (Checkland 1972). He sees Forster's definition of AR (augmenting that of Rapoport) as fitting the Lancaster systems projects:

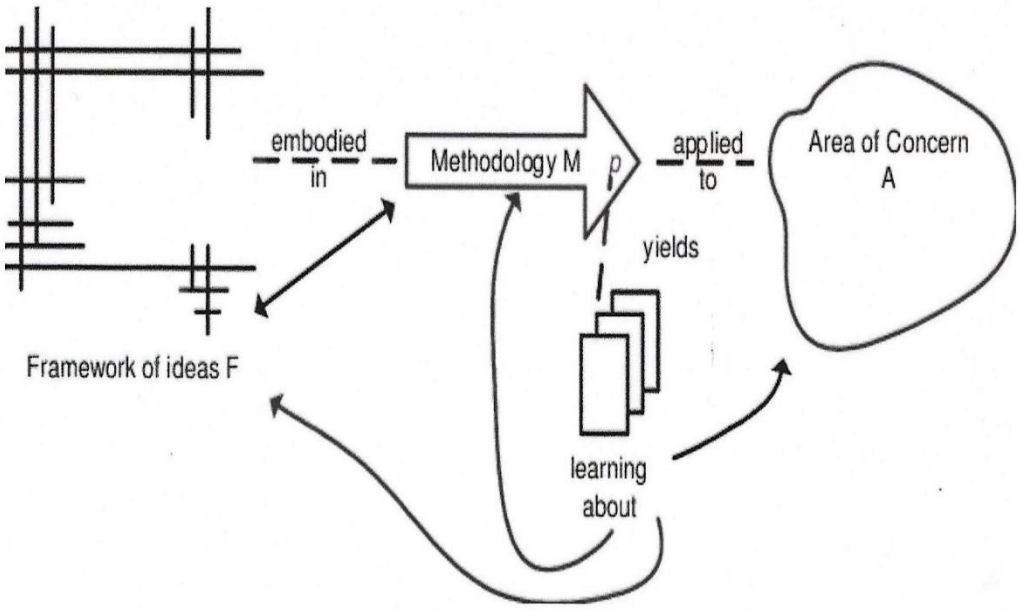
> A type of applied social research differing from other varieties in the immediacy of the researcher's involvement in the action process and the intention of the parties, although with different roles, to be involved in a change process of the system itself. It aims to contribute both to the practical concerns of people in an immediate problematic situation and to goals of social science by joint collaboration within a mutually acceptable ethical framework (Forster, quoted in Checkland 1972, p. 57).

Through Foster, he identifies Lewin as the father of AR, approving of Lewin's objection, deriving from the Gestalt psychologists, to breaking apart complex social events to study them in a laboratory. *Systems Thinking, Systems Practice* (Checkland 1981) contains an extended

account of AR with the Forster definition still central. *Soft Systems Methodology in Action* (Checkland and Scholes, 1990), in Checkland's view, illustrates this mode of AR throughout (Checkland 1999, p. A39).

The FMA framework, as a means of understanding the nature of research of all kinds, was first introduced by Checkland (1985a) and later transferred by Checkland and Holwell (1998a, 1998b) to the specific context of AR. Figure 1 shows the Checkland and Holwell representation. Translated into the context of AR, the action researcher

Figure 1. The FMA framework of elements relevant to any piece of research (Checkland and Holwell, 1998b, p.23)



first joins a real-world problem situation, an *area of concern* (A) that is of interest to them and those involved. It is deemed essential, for the advancement of research, that they declare the *framework of ideas* (F) and associated *methodology* (M) they are about to use in trying to secure improvements. They take part in action in the A and reflect on what has happened using the F and M. The FMA framework helps ensure improvements from the point of view of participants in the problem situation and enhancements to the F and M to increase their usefulness in the future.

Checkland and Holwell use this framework to comment further on the nature of AR. For them, AR assumes that social reality, unlike physical reality, is in a state of constant flux. No two social situations can be assumed to be the same, so the results of research cannot be justified based on the natural scientific criterion of repeatability. However, Checkland and Holwell (1998a, p. 20) argue, an appropriate "truth claim" is still available to AR if the researchers declare their F and M in advance and keep careful records of what exactly occurs in A. The outcomes can then be related back and described in terms of the F and M. This allows the *recoverability* of the whole research story by interested outsiders and enables productive debate about the findings. Checkland and Holwell are critical of much AR for failing to insist that the F is declared in advance because this makes it impossible to explain how the findings are arrived at and to learn more about that F:

> The literature of AR has so far shown an inadequate appreciation of the need for a declared epistemology and hence a recoverable research process (Checkland and Holwell 1998a, p. 20).

For Checkland and Holwell, the success of the Lancaster AR programme can be understood in terms of its adherence to the FMA framework in over 300 projects over a 30-year period. Being upfront about the F and M ensured that action taken to bring about improvements, from the point of view of participants in the problem situation, could produce learning about the F and M and guide refinements that would make them fitter for purpose in future projects. A positive feedback loop between theory and practice was generated. The next section looks at how this played out. It will enable us to assess, in the section after, whether all was quite as positive as has been painted.

The Development of Soft Systems Methodology (SSM) through the Lens of the FMA Framework

The development of SSM is now considered from its origins in Systems Engineering to its current 'mature' state. This is done in terms of five transitions marked by SE and the four different standard representations of the methodology. Each transition is detailed by setting out the M, F and A at the beginning of the relevant research period and then considering what learning took place that led to the transition to the next F, M and A.

a. *From Systems Engineering (SE) to Blocks and Arrows*

The SE methodology (M) first used in the AR programme is expounded by Jenkins in his inaugural lecture 'Systems and their Optimisation':

> Systems engineering is the study of complex systems in their totality so as to ensure that they achieve their overall objectives as efficiently as possible (Jenkins 1967, p. 6).

The systems engineer should identify the problem and the system in which it is located, determine the objective of the system and of the wider system, model the system, optimize it with respect to the relevant economic criterion, establish control procedures, and ensure proper implementation. Jenkins believed that using this M would lead to the discovery of generic models of functional systems, such as purchasing, production, marketing, distribution, finance, human resources, etc. Brian Wilson, an electrical and control engineer who joined Jenkins in the department in 1966, continued to pursue this agenda – as we shall see.

Checkland's own inaugural lecture, 'Systems and Science, Industry and Innovation', endorsed the SE of Hall (1962) and Jenkins as the starting point for the Lancaster research:

> Having defined our problem and its relevant system, and having defined the system's objectives with regard to constraints from the wider system, the Systems Engineer can begin to design an improved system (Checkland 1969, p. 14).

He does raise concerns, supported by a reading of Churchman's book *Challenge to Reason*, that the rationality of SE might undermine human values and threaten creativity. But these are not developed. Nearly 30

years later, in a Valedictory Lecture given at Lancaster, Checkland suggests that his wife, Glenys, would have accurately described his inaugural as

> … a naïve Inaugural in which you indicated that you would try to bring the human dimension into Systems Engineering (Checkland 1997, p. 3).

He was, at least, clear that SE applied to social systems would face considerable difficulties. This proved to be the case and Checkland, more open than Jenkins to questioning the appropriateness of SE outside the technical field, rapidly concluded that the M itself would have to be radically changed to render it fit for dealing with the greater complexity exhibited in the managerial context.

The F underpinning SE is explained by Checkland (1999). The assumption is that systems exist in the world, their objectives can be clearly defined, and they can be engineered to achieve those objectives. This orientation he sees as deriving from the 'goal-seeking' approach to management found in the work of Herbert Simon. Organizations, following Optner, are seen as systems with functional sub-systems. It is possible to acquire knowledge about how they and their sub-systems work and to optimize their performance.

Checkland appreciated Jenkins's broad understanding of 'engineering' and his desire to extend the *areas of concern* (As) of SE. Originally appointed as a Professor of Commercial Systems, he determined to apply the approach to "problematical management situations … large and small, public and private" (Checkland 2011, p.498).

Four early engagements helped steer Checkland into breaking with SE. One involved work on the Anglo-French Concorde project. From an SE point of view, the project seemed to have a clear objective - to put together a structured set of activities to create the specified supersonic jet aircraft. But there were competing political objectives, e.g. to manage relations between the British and French Governments, which limited progress on that technical objective (Checkland 1999). In the other three, in the textile, engineering, and publishing/printing

industries, the main difficulty was that the situation of concern was so vague that the clients were unable to specify the nature of what they wanted to achieve. In a textile firm there were at least a dozen plausible candidates for the role of 'the problem' (Checkland 1972).

From these and other projects, learning was accumulated which informed the first *Blocks and Arrows* version of SSM. Indeed, three of the four 'key thoughts' that Checkland came to see as central to SSM saw the light of day at this time (Checkland 1999). First, it is more useful to see problem situations as made up of people trying to take purposeful action that has meaning for them than as hierarchies of systems in need of optimisation. To express this insight, relevant 'human activity system' models can be constructed. Checkland regarded the delineation of this type of model, capturing the activities necessary to generate purposeful action, as one of the most important breakthroughs in the development of SSM. Second, given the complexity of human affairs, many human activity system models will likely be relevant and insightful. SSM, therefore, should proceed by building several such models, in each case being clear about the worldview, or *Weltanschauung*, that makes it meaningful. Third, while the models produced by SE are meant to be models of real world systems or blueprints for design, human activity system models are contributions to discussion and debate. Participants use the models as part of a learning process that leads to decisions about what changes are feasible and desirable.

As well as learning from the experience gained in projects, Checkland was helped in transitioning to SSM by reflecting on ideas developed by thinkers who did not share the F underpinning SE. He talks of the importance of a one-week course he took with C. West Churchman, just 6 weeks after starting at Lancaster, in which Churchman presented the material in his books *Challenge to Reason* and *The Systems Approach.* Checkland was impressed by the rigour with which Churchman, and his doctoral student Ackoff, sought to translate philosophical ideas to the world of affairs and began to see both as mentors. The most important concept he took away from the

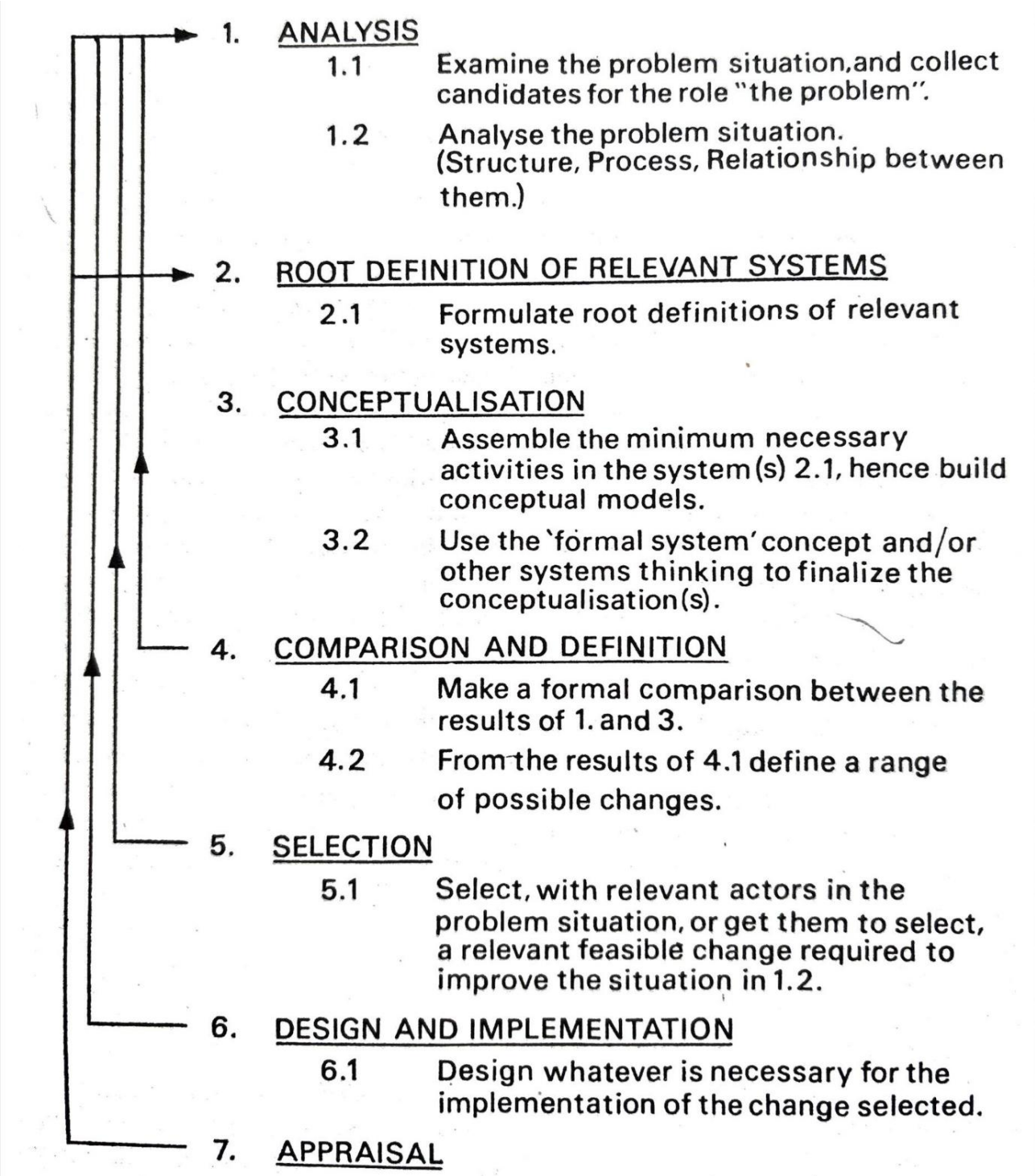
course was that of *Weltanschauung* – a particular perspective on the world. Meanwhile, Ackoff's (1971) early attempt to bring clarity to systems terms and concepts (later pursued, with Emery, in the book *On Purposeful Systems*) seems to have helped him formulate the idea of systemic models of purposeful activity. Checkland credits the concept of *Weltanschauung* and the idea of purposeful activity models as "providing a path which led us from Systems Engineering to SSM" (Checkland 2011, p. 503). Another decisive step in the direction of SSM in its *Blocks and Arrows* format stemmed from his engagement with the work of Sir Geoffrey Vickers which, he says, "has had by far the greatest impact on the development of SSM" (Checkland 2011, p. 507).

b. From Blocks and Arrows to Seven-Stages

Only three years after joining the department at Lancaster, Checkland was able to present a first articulation of SSM in a paper titled 'Towards a systems-based methodology for real-world problem solving' (Checkland 1972). For those of us who studied at Lancaster during the 1970s this was taught as Systems Engineering Methodology 2, in contrast to Jenkins's Systems Engineering Methodology 1 (Checkland 1977; Mingers 2000). It is referred to here, following Checkland (1999), as the *Blocks and Arrows* version of SSM.

The methodology is shown as having seven steps, as in Figure 2. All the lessons from the early AR projects are incorporated. 'Analysis' is no longer conducted 'in systems terms'. The search for systems to engineer, and the requirement for clearly defined objectives, are abandoned. Instead, the aim is to build up 'the richest possible picture'

Figure 2. The 1972 version of the methodology (Checkland, 1972, p. 73)



of the problem situation by viewing it in terms of 'structure', 'process' and 'climate' (the relationship between structure and process). Guidelines are provided setting out how this can be done. There is no

indication, at this time, that a pictorial 'rich picture' might be useful. From this examination, it becomes possible to select some notional systems, reflecting different viewpoints, which are relevant to improving the problem situation. Checkland provides the example of a church which might reasonably be conceived of as 'a social welfare system', 'a ritual-organising system', 'a belief-maintenance system', 'a system to provide support in the face of unanswerable questions', etc. 'Root definitions', concise descriptions of the relevant systems, are formulated. 'Conceptual models' of what is implied in the root definitions are then constructed. These should contain "in correct sequence the *minimum* activities which are *necessary* in the human activity system described by the root definition" (Checkland 1972, p. 64; italics in the original). They consist of verb statements arranged in a logical order. Conceptual models are to be checked against a 'formal system' concept – a generalised model of any human activity system derived from Jenkins' thinking and Churchman's 'Anatomy of System Teleology' – to test whether they are 'not basically deficient'. They are then employed, in a formal comparison with what was learned from 'analysis', to suggest possible changes (structural, procedural, attitudinal, or environmental) that would improve the situation under investigation. Feasible changes are selected with actors in the problem situation, implemented, and the results appraised.

Looking back at the F underpinning 1972 SSM, Checkland (1999) sees that it is still imbued with the legacy of 'hard systems thinking'. It has "a whiff of Systems Engineering about it" (Checkland 2011, p. 503). The implication remains that 'hard systems' and 'soft systems' exist in the world and the language of problem situations has not fully replaced that of real world problems. Further, the straight arrows and rectangular blocks in the diagrams and models speak of a systematic rather than systemic orientation. Nevertheless, the focus on implementing change rather than designing an efficient system signals significant progress. Churchman's thinking is referenced as providing inspiration for the direction in which SSM was heading. Emery and Trist's work on 'turbulent fields' receives a passing note. The major

input to the F, however, comes from Checkland's reading of the work of Sir Geoffrey Vickers. He sees Vickers's books as pointing the way forward from the 'goal-seeking model' of human behaviour, which dominates traditional management science, towards one of 'maintaining ongoing relationships'. 'Problems' are seen to exist in the context of the 'appreciative systems' different participants in a problem situation bring to the table. Making progress in dealing with problems depends on ensuring that appreciative systems are sufficiently shared for changes to be agreed upon. Checkland had begun to see SSM as a way of managing this process.

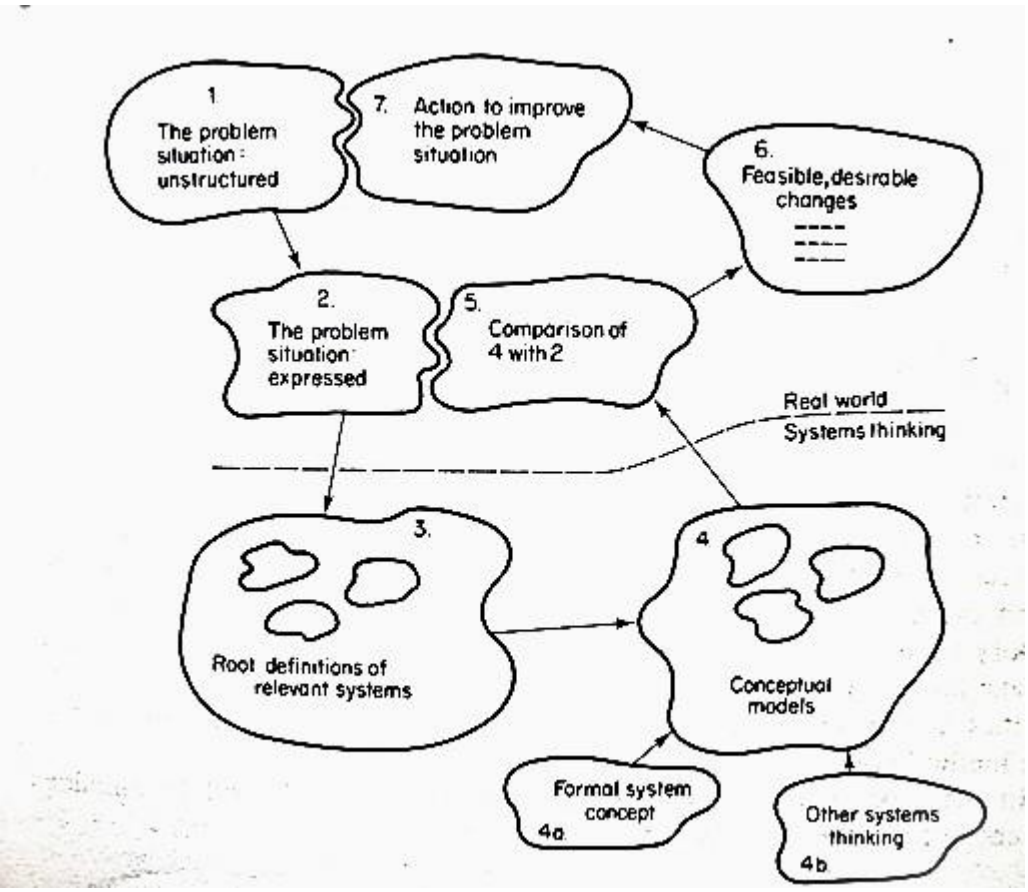
During the next decade, leading up to the publication of *Systems Thinking, Systems Practice* (Checkland 1981), the AR programme that had resulted in the early development of SSM was rigorously pursued in many different As *(areas of concern)*. Six systems studies are identified by Checkland (1981) as having significant impact. Because of the learning obtained from these, the way the methodology was displayed began to change. By 1975 SSM was being presented in its *Seven-Stages* format (see Figure 3) as "a cluster of seven activities in a circular learning process" (Checkland 1999, p. A12). Experience in using the techniques designed to put SSM into practice led to significant developments. In 1977, Naughton, confronted with the task of teaching SSM at the Open University, found it necessary to identify the Constitutive Rules that had to be obeyed in any SSM study. Checkland (1981) was able to make use of this elaboration.

During the period 1972 to 1981, there were strides forward in the F upon which SSM rested. Checkland gave more thought to the importance of the concept of *Weltanschauung*. He delved deeply into 'related work' in the systems tradition. Of even more significance, he broadened his reading in the social sciences and started asking questions about the vision of social reality implied by SSM

c. From Seven-Stages to Two Streams

By the time *Systems Thinking, Systems Practice* was published, in 1981, the *Seven-Stages* version of SSM was firmly in place. It is shown in Figure 3.

Figure 3. SSM as Seven-Stages (Checkland, 1981, p. 163)



The aim of the new representation was to emphasise the 'systemic' nature of the methodology. The circular form conveys more clearly that the stages need not be followed in a rigid engineering-like sequence. It is possible to start anywhere, and the activities will mutually inform one another in a manner that encourages continuous iteration around the cycle as more learning is obtained. Checkland (1999) believes that this new version looks more 'organic', 'human', 'natural' than the 1972

description. The ‘certainty’ of achieving fixed outcomes, indicated by rectangular boxes, is gone. The ‘fried-egg shapes’ convey that work is in progress on a unique application. The other obvious new feature in the *Seven-Stages* version is the line dividing stages 1,2,5,6 and 7, the ‘real world’ activities that must involve people in the problem-situation, from stages 3,4,4a and 4b, ‘the ‘systems thinking world’ which may or may not involve them. Its purpose is to emphasise the shift that takes place in a SSM study, only hinted at in 1972, from the everyday language of the problem-situation to the ‘meta-language’ of systems (Checkland, 1981).

Considerable effort, in the 1970s, went into refining methods for accomplishing the different stages of the M, and these are all incorporated in the 1981 *Seven-Stages* description. Stages 1 and 2 seek to ‘express’ the unstructured problem situation by building up ‘the richest possible picture’ of that situation. This ‘neutral’ description should allow a range of relevant, but different, perceptions of the problem situation to come to the fore. The idea that any form of ‘analysis’ in systems terms is taking place is abandoned. The concepts of ‘structure’, ‘process’ and ‘climate’ are, however, seen by now as ‘well-tested’ guidelines for expressing the problem situation. These can be expanded by considering the ‘social systems’, consisting of ‘roles’, ‘norms’ and ‘values’, and ‘political systems’, concerning the exercise of power, that they constitute.

Stage 3, the construction of ‘root definitions’ of relevant systems, receives considerable attention. It is emphasised that each definition will express a particular *Weltanschauung* and a section of *Systems Thinking, Systems Practice* is devoted to exploring this concept and its importance in the development of SSM. The emergence of the CATWOE (Customers, Actors, Transformation, Weltanschauung, Ownership, Environmental Constraints) mnemonic, for ensuring root definitions are ‘well-formed’, is described at length. It was clear to Smyth and Checkland (1976) that root definitions needed to reflect the characteristics of the ‘formal system model’. And once this was obvious, it became possible to discern how previous SSM studies

would have been enriched by paying attention to all six elements of CATWOE. Similarly, the distinction between 'primary task' and 'issue-based' root definitions, recognised by Checkland and Wilson (1980), could be justified by looking back at how it could have enhanced earlier projects. Primary task root definitions are those related to the explicit task of an organisation; issue-based definitions pertain to specific issues at a point in time and frequently cut across organizational boundaries. Checkland notices that the concept of 'primary task' was well-established in the literature of socio-technical systems thinking but still insists that it was discovered experientially by the Lancaster researchers as part of their action research program. Reading Rice's (1963) and Miller and Rice's (1967) accounts of the evolution of the concept, it is reasonable to suggest that Checkland absorbed more of the socio-technical work than he thinks. Both books assert that 'primary task' can be used as a heuristic concept to explore different organisational models, possible conflict between them, and whether compromise is necessary. Rice states:

> From different assumptions about the environment, and the changing relations between enterprise and environment, models based on different import-conversion-export relationships can be constructed. The models can then be compared with each other and with the existing organization (Rice 1963, p. 17).

Rice and Miller and Rice even use the example of different visions of the UK prison service - as a punishment system, a confinement system and a rehabilitation system - to explain their thinking. This is, of course, akin to Checkland's pop-festival case, used again in *Systems Thinking, Systems Practice*, and has become a staple illustrative example among soft systems thinkers. Checkland is correct, however, in his claim that SSM offers greater clarity than socio-technical thinking in demarcating root definitions as concise representations of perceptions of the world rather than actual models of reality.

Stage 4 requires the making and testing of 'conceptual models'. Each model describes the purposeful human activities implied by one of the root definitions. They are certainly not models of part of the real world. As in the 1972 version of the M, Checkland insists that a model contain the minimum number of verbs, arranged in a logical order,

necessary to achieve the transformation named in the root definition. According to 'the law of model building' the verbs must specify activities that actors can directly carry out. The activities will involve acquiring inputs, transforming them, and doing something with the outputs. More attention, however, is now given to activities concerned with the measures of success relevant to each model and with monitoring whether they are met. Once the models have been built, again as with 1972, they can be tested against what is now called the 'formal system model', a model of what any human activity system must be like, to check that they are not fundamentally inadequate. Other systems thinking 'which the analyst reveres', such as socio-technical thinking, Forrester's 'system dynamics' or Beer's 'viable system model', can also be 'mined' at this stage to 'extend or recast' a conceptual model. Models can be taken to higher resolution levels by expanding activities into sub-systems, each built and tested according to the same general rules employed in constructing root definitions and conceptual models. It is essential in doing this to obey 'the law of conceptualisation' which insists that a subsidiary system cannot be defined and modelled until the system served is defined and modelled. Information systems provide an example of when this makes obvious sense, but where such sense rarely prevails.

The 'what' (you would do) activities set out in the conceptual models can then, at Stage 5, be taken above the line, back into the 'real-world', to debate 'how', if at all, they are carried out and 'how' they might possibly be done better. The *Seven-Stages* version of the M provides four different ways of achieving this. Finally, at Stages 6 and 7, the comparison of the models with 'reality' is used to engender debate about possible structural, procedural and attitudinal changes. Those changes, agreed with people in the problem situation, will be both 'systemically desirable', having emerged from the root definitions and conceptual models, and 'feasible', given the characteristics of the situation.

In *Systems Thinking, Systems Practice* (1981, p. 253), Checkland further clarifies the *Seven-Stages* M by offering a slightly modified set

of Naughton's Constitutive Rules which 'must be obeyed' to carry out an inquiry of a soft systems nature. In effect, these do little more than restate the 7-stage process, while relegating certain techniques, such as the use of 'structure, process and climate', during the expression stage, to 'Strategic Rules' which an analyst may or may not find helpful.

Attention is also devoted, in that book, to what 'a system to use the methodology' would look like. Reflections on the roles of 'client', 'decision taker', 'problem owner' and 'problem solver', in any problem situation, lead to the specification of 'A Workbook for Starting Systems Studies'. This requires the identification of the client and his (sic) aspirations, reflection on the 'problem-content' system, in terms of the decision-taker's/problem owner's understanding of the problem, and consideration of the problem-solving system, particularly the resources it can command.

Finally, it is worth stating that, although the methodology underwent further changes of shape, Checkland continued to see merit in the *Seven-Stages* version as a pedagogical device – it is easy to understand, unfolds logically, and contains the right number of concepts according to Miller's work on the 'magical number 7'. He was still content using it to explain an application of SSM published in 1985 (Checkland 1985b).

We now turn to the F underpinning the *Seven-Stages* M. In *Systems Thinking, Systems Practice,* Checkland offers his most comprehensive account of the F supporting SSM, taking time to consider both 'related work elsewhere', 'the nature of social reality', and 'the social reality implied by soft systems methodology'. In discussing 'related work elsewhere', he distinguishes his systems thinking from socio-technical systems thinking and the work of Ackoff, Ozbekhan, and Churchman. The argument is the same in each case. Even Churchman, whose work on 'enquiring systems' he particularly respects and makes use of, fails to fully escape the constraints of the goal-seeking paradigm underpinning hard systems thinking. Only Vickers succeeds in seeing beyond this paradigm and developing an alternative, based on 'establishing and modifying relationships through time'. In

Checkland's view, this corresponds much more closely to what typically goes on in organisations:

> Vickers suggests replacing the goal-seeking and goal-seeking-with-feedback (cybernetic) models by one in which personal, institutional or cultural activity consists in maintaining desired relationships and eluding undesired ones (Checkland 1981, p. 262).

To maintain desired relationships in a situation, those involved need to expose the 'appreciative systems' they use to selectively view reality, and make judgements about it, to a learning process which seeks to bring about a degree of alignment. Checkland regards his M as being well equipped to guide such a learning process:

> In a sentence, the work of Churchman and Vickers can be related to soft systems methodology in the following way: the methodology is a Singerian enquiring system whose mode of operation provides a formal means of initiating and consciously reflecting upon the social process of 'appreciation' (Checkland 1981, p. 264).

Transitioning to a discussion of 'the nature of social reality', Checkland insists that the evolution of the M was not based on any explicit theory. Indeed, he says, it is the absence of "generally accepted and well-tested models of social and political systems" (Checkland, 1981, p. 230) that prevents the development of the expression stage beyond the relatively rudimentary concepts of roles, norms, and values. Nevertheless, Checkland can identify a tradition of social theory to which his work closely relates. In general, the M echoes the concern of the 'Weberian tradition', with its emphasis on the subjective understanding of 'meaning', rather than the 'Durkheimian tradition', which apes the natural sciences in giving prominence to the objective study of social facts. Within the 'interpretive' strand of social theory initiated by Weber, it is the 'phenomenology' of Husserl and Schutz and the 'hermeneutics' of Dilthey that Checkland sees as exhibiting the closest similarities to Vickers's formulations and SSM:

> In summary, we have in phenomenology and hermeneutics an attitude towards social science which takes as its prime datum not the world external to observers of it, but the observer's mental processes …human beings in the social process are constantly creating the social world in interaction with others. They are negotiating their

> interpretations of reality, those multiple interpretations at the same time constituting the reality itself (Checkland 1981, p. 277).

Taking the argument a stage further, Checkland makes use of work by Burrell and Morgan (contemporaries for a period at Lancaster University) to distinguish the social theory underpinning his M to that implicit in other systems approaches. Burrell and Morgan identify four sociological paradigms depending on whether they emphasise objective or subjective aspects of social reality and privilege regulation or radical change. These are 'functionalist' (objective and regulation), 'interpretive' (subjective and regulation), 'radical structuralist' (objective and radical change), and 'radical humanist' (subjective and radical change). For Checkland, social system theory and methodologies such as systems engineering, systems analysis, and formal OR are clearly functionalist in nature. Whereas:

> The social theory implicit in soft systems methodology … would lie in the left-hand quadrants with hermeneutics and phenomenology, although the position would be not too far left of the centre line because the methodology will over a period time yield a picture of the common structurings which characterize the social collectivities within which it works. Also, given the analyst's complete freedom to select relevant systems [which could express the need for radical change] … the area occupied must include some of the 'subjective/radical' quadrant (Checkland 1999, pp 280-281).

A discussion, drawing on Mingers (1980), seeks to justify the extension of the M's competence into the subjective/radical quadrant by comparing the emancipatory potential of SSM with Habermas's 'radical reformism'. In Checkland's opinion (1981, p.265), the very success of the M offers support for the phenomenological/ hermeneutic account of the nature of social reality.

The decade following the full articulation of the *Seven-Stages* version of the M, in *Systems Thinking Systems Practice*, saw its use in AR in a wide range of As – in industry, the civil service, the NHS, in projects large and small, lasting a few hours to more than a year (Checkland and Scholes 1990; Checkland 1999, p. A15). Ten of these studies are discussed in detail in *Soft Systems Methodology in Actio*n (Checkland and Scholes 1990). Checkland is clear about the nature of

the learning obtained and how it led to the formulation of the next *Two Streams* representation of SSM:

> So, although a debate structured by questioning perceptions of the real situation by means of purposeful activity models was always insightful, moving on to action entailed broader considerations (Checkland 1999, p. A14).

These 'broader considerations' concerned how to 'read' the history, culture and politics of situations so that a better understanding could be obtained of what accommodations between conflicting interests might be possible. This would enable the intervention to be conducted in a more informed manner and increase the likelihood of achieving feasible and desirable change. The matter was seen to be of such importance that it necessitated the addition of a 'cultural and political stream' of analysis to run concurrently with the 'logic-based stream'. A dramatic change in the 'shape' of the M was required, the old *Seven-Stages* version now seen as having

> …a rather mechanistic flavour [that] can give the false impression that SSM is a prescriptive process which has to be followed systematically (Checkland 1999, p. A9).

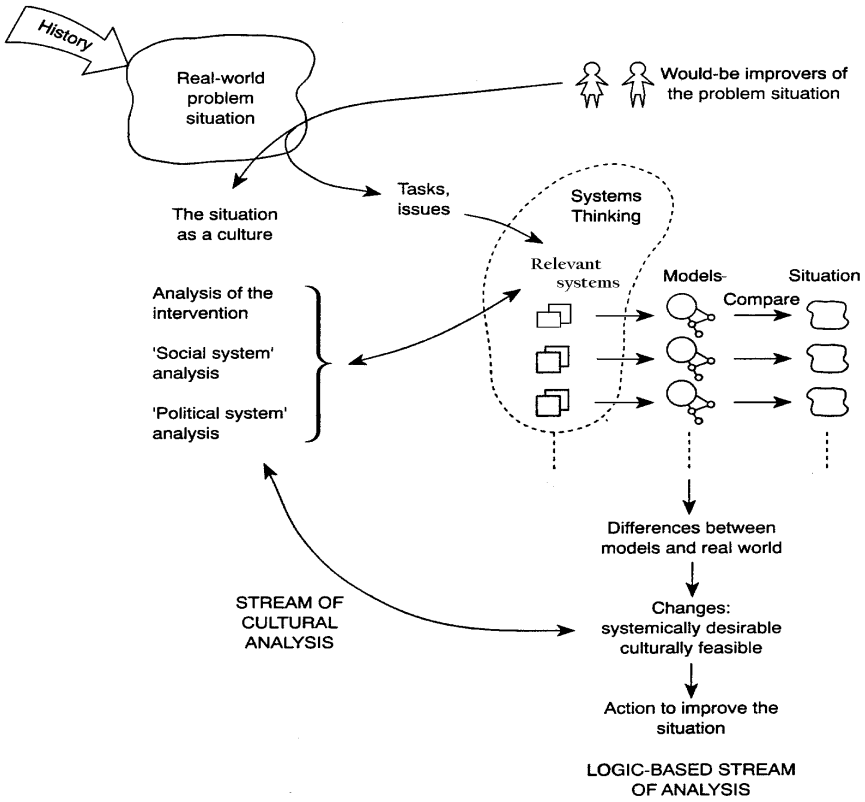
It is difficult to see the attention given to politics and power, in the *Two Streams* version of the M, as anything but an attempt by Checkland to respond to the many criticisms launched at SSM's deficiencies in this respect in the 1980s (Jackson 1982; Mingers 1984; Rosenhead 1984; Wilmott, 1989). But he does not acknowledge this.

The *Two Streams* model also benefited from the clarity achieved by Checkland and colleagues, "as experience accumulated" in the 1980s, around the precise definition of 'hard' and 'soft' systems thinking (Checkland 1999, p. A10). This increasing clarity can be traced in Checkland's publications of the time (1983, 1985a). The *Two Streams* version of the M was first expounded by Checkland, in 1987, in a plenary at the annual meeting of the International Society for General Systems Research and is fully set out in *Soft Systems Methodology in Action* (Checkland and Scholes 1990).

d. From Two Streams to Four Main Activities

The *Two Streams* representation of SSM is shown in Figure 4.

Figure 4. The Two Streams version of SSM (Checkland and Scholes, 1990, p. 29)



Would be improvers of a problem situation begin an intervention aware that the history of that situation will influence what can be achieved just as much as any logic-based systems work. It is therefore helpful to conduct two streams of enquiry – a stream of cultural analysis and a logic-based stream of analysis. These interact and inform one another throughout the study. For example, knowledge about the culture will suggest what relevant systems can be usefully pursued, while the systems seen as relevant by those in the problem situation will provide learning about the culture. It is this close interaction between the world of the problem situation and the systems thinking world which explains the demise of the dividing line between the two

present in the *Seven-Stages* version. In Checkland's words "it had served its heuristic purpose" (1999, p. A15). The aim of the M is stated as arriving, through structured debate, at an accommodation between different worldviews and interests that will allow some feasible and desirable changes to be implemented. This must be seen as a temporary phase in the operationalisation of Singer's never-ending cycle of learning.

Examining the cultural stream of analysis, we find in *Soft Systems Methodology in Action* (Checkland and Scholes 1990) the first discussion of the usefulness of pictorially representing problem situations in 'rich pictures'. This will be a surprise to many who view the technique as central to SSM and assume it was there from the start. Checkland and Scholes refer readers to various examples of rich pictures in the book and comment on the

> … preoccupations of compilers of rich pictures: expressing relationships and value judgements; finding symbols to convey the correct 'feel' of the situation; indicating that many relevant relationships preclude instant solutions (Checkland and Scholes 1990, p. 45).

They hold back from offering formal advice on how rich pictures should be drawn or what they should look like. It is noteworthy that the orthodox use of the concepts of structure, process, and climate at the 'expression' stage does not survive in the *Two Streams* version of the M.

The cultural stream of analysis then proceeds through Analysis One - 'analysis of the intervention'; Analysis Two - 'social system analysis'; and Analysis Three - 'political system analysis'. Analysis One reviews the roles of 'client', 'would be problem-solver', and 'problem-owners'. Checkland and Scholes emphasize that the 'problem-solver' will always have many possibilities to choose from for the role of 'problem-owners' and this can provide inspiration for insightful relevant systems. Analysis Two seeks to chart the ever-changing interaction between 'roles' (both formal and informal), 'norms' (expected behaviours in roles) and 'values' (judgements of performance in roles) in the problem situation. Analysis Three concerns itself with politics in the sense of

how power is expressed in the problem situation and can influence relations between different interested parties. It draws on work by Frank Stowell (1989a) and asks what 'commodities' or 'embodiments' of power – formal authority, intellectual authority, charisma, reputation, access to information, etc. etc. – exist, are employed, and are significant in the problem situation. Checkland quotes Stowell on how the metaphor 'commodity' is used in this context:

> 'Commodity', then, is the proposed means of providing organizational members involved in change with a practical means of addressing power. Acknowledging, with Giddens, 'that speech and language provide us with useful clues as to how to conceptualise processes of social production and reproduction', what has been suggested within this thesis is an idea by which the notion of power can be articulated in terms which are appropriate to a given organizational culture and which can be understood by those most affected (Stowell 1989a, quoted in Checkland 1999, p. A20).

Checkland and Scholes are aware that open discussions about the use of power are fraught with difficulties and counsel SSM users to be circumspect when carrying out Analysis Three as part of the cultural analysis.

The logic-based stream of analysis sees some additions and modifications to the procedures described in the *Seven-Stages* version of the M. Checkland and Scholes suggest that it is useful to think of a 'full' root definition's core transformation as 'a system to do X by Y in order to achieve Z' (letters later replaced by P, Q, and R). This supports CATWOE in ensuring the thinking covers the system, sub-system and wider-system levels. Checkland suggests that referring to purposeful activity models as 'holons' will help prevent people from confusing them with models of the real world – an idea he later admits did not catch on. Further thought is given to how 'measures of performance' should be expressed in these models. Checkland and Scholes argue that paying attention to '3Es' – efficacy (do the means work?), efficiency (is resource use kept to the minimum?), and effectiveness (is the longer-term aim met?) - is always relevant and can add richness to the comparison stage. In certain cases, it is also useful to consider the criteria of ethicality and elegance. Use of the 'formal system model' is finally dropped, its purpose seen as fulfilled by CATWOE. In any case,

its existence tended to suggest that models were to be used for design rather than for structuring debate. Finally, there is discussion of the use of SSM as a vehicle for implementing whatever feasible and desirable changes are agreed.

There are three further developments in the conceptualization of the M that Checkland and Scholes (1990, p. 298) are keen to emphasize. These are a Mode 1 – Mode 2 distinction, further thought on 'the system to use SSM', and a new set of Constitutive Rules. Much of the early work on SSM, they argue, concentrated on developing the methodology. It was introduced as an external recipe to drive an intervention in a structured and sometimes sequential manner. This is now defined as Mode 1 usage. As experience and confidence grew, those using SSM began to internalize the approach. Once this happens it becomes possible to remain much more 'situation-driven'. The M ceases to dominate what is done and becomes instead the basis for reflecting on what is happening during the everyday flux of events. In keeping with the orientation of the *Two Streams* version, it is used flexibly and only occasionally breaks the surface to interact with ongoing ideas and affairs. This is Mode 2 SSM.

The 'system to use SSM' is now modified to cover the different possible uses of the M, whether they are near-Mode 1 or near-Mode 2 applications. It also needs to incorporate new developments such as the 3Es, which enable it to adapt as events change during the intervention. A new version of a 'system to use SSM' is presented which is based on the work of Vickers and is said to map all the experiences of using the M.

Finally, the Constitutive Rules are rethought to correspond more closely to the *Two Streams* expression of the M and both Mode 1 and 2 ways of using it. Checkland says of the new rules:

> These defined five characteristics of uses of SSM and set out its epistemology (rich pictures, CATWOE, etc.). A use of SSM was one which could be described using these concepts and language (Checkland 1999, p. A34).

The epistemology said to make sense of the Rules is that the real world is seen as "the unfolding interacting flux of events and ideas experienced as everyday life" (Checkland and Scholes 1990, p. 288).

There is little in these Constitutive Rules to excite those who might have expected the radical changes in the M signaled by the *Two Streams* model to be accompanied by interesting changes to the F. *Soft Systems Methodology in Action* simply offers further insight into the nature of a largely unchanging F. Building on earlier work in the 1980s (Checkland, 1983, 1985a), Checkland and Scholes can clearly differentiate the 'hard' and 'soft' traditions in systems thinking. The former, of which Systems Engineering is an example, assumes that the world is systemic, contains systems, and that their performance can be optimized using a systematic M. The latter treats reality as problematical, as unknowable ontologically, but argues that the process of engaging with it can be conducted using a systemic M. SSM, for example, embodies a systemic process of learning, a cyclic course of enquiry, which happens to use systems models as part of that process. In the shift from 'hard' to 'soft', systemicity is transferred "*from the world to the process of inquiry into the world*" (Checkland 1983, p. 672; italics in the original). Also reinforced is the idea, first developed in Checkland (1985a), that 'soft' systems thinking has to be the general case in human affairs and 'hard' systems thinking is the occasional special case when there is consensus on what needs doing, Otherwise, there is reflection on how SSM provides a M that embodies Schon's notion of 'reflection in action' and transcends it by allowing analysis and development of the capacity for problem solving. And a suggestion, in a statement best forgotten, that there is an element of postmodernism in SSM because the interpretations of the world with which it works have 'no absolute or universal status'.

Later, Checkland (1999, pp. A19-20) takes more time to consider the relationship between the F and M in the *Two Streams* model. He suggests that to make sense of Analysis Two and Three you must accept the view "that social reality … is to be seen as continuously socially constructed by individuals and groups (the latter never perfectly

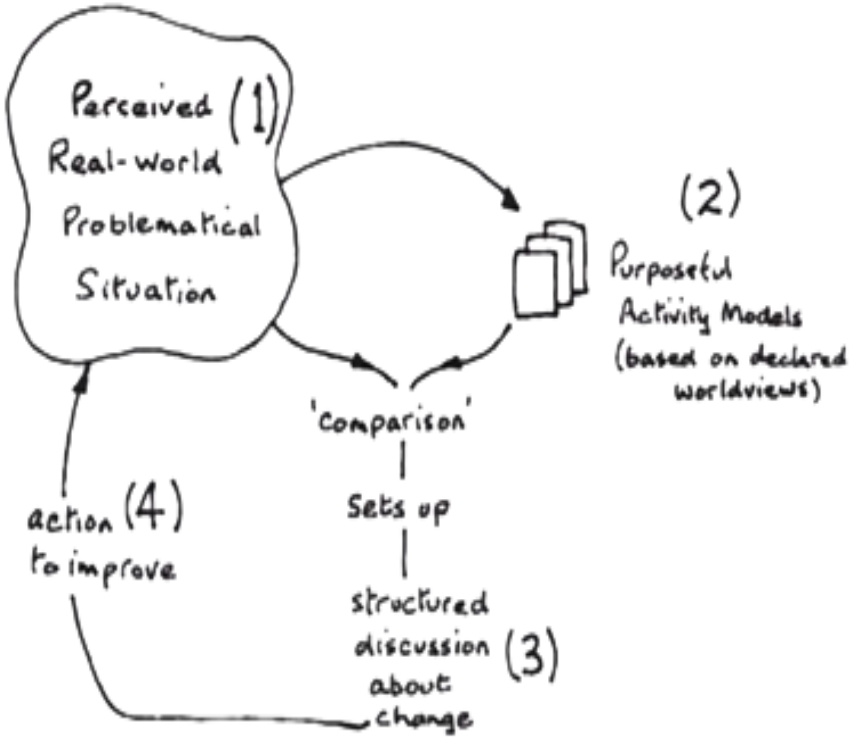
coherent)". He relates this stance to the theoretical contributions of Weber, Schutz, and Husserl, and its practical realization in the M to the "autopoietic model teased out of the work of Vickers on appreciative systems" by himself and Casar. Analysis Three is said to move "beyond the model of an appreciative system" but to be compatible with it. This is intelligible if we consider what Stowell sought to achieve by exploring organizational power through the metaphor of 'commodity' (Stowell, 1989b). Stowell sees the metaphor as related to bartering and as capable of providing insight into the use of power in different enterprises according to what is available for bartering. It thus provides a very practical tool for understanding how power is exercised in any organizational setting. Checkland picks up on this, insisting that he is not interested in the question 'What is power?', but instead wants an idea that can open up discussion on the often taboo subject of how power is manifested, distributed and used in organizations. Thinking of power as a commodity enables such discussion. It fits with the interpretive rather than positivist or radical way of understanding power (Stowell 2021a).

In the 1990s following the publication of *Soft Systems Methodology in Action*, in which the *Two Streams* model was fully articulated, the influence of SSM 'spread both geographically and by discipline' (Mingers, 2000). Checkland and Scholes (1990) had already indicated that 'information systems' was a field to which SSM was well suited and Checkland's next book, this time with Holwell, *Information, Systems and Information Systems* (Checkland and Holwell 1998b) was devoted to conceptual clarification in this area of concern and describing some relevant experiences using SSM. Significantly, Checkland (1999, p. A9) sees the idea that "models of purposeful activity can provide an entry to work on information systems" as the last of the 'four key thoughts' that shaped the development of SSM. In the 1999 book, it is yet another representation of SSM that is given priority – the *Four Main Activities* model. It is to this which we now turn.

e. Four Main Activities and Beyond

In *Soft Systems Methodology in Action*, alongside the *Two Streams* model, Checkland and Scholes offer an alternative *Four Main Activities* version of SSM to explain its basic shape. Checkland (1999, p. A15) later explains that even the *Two Streams* model "was felt to carry a more formal air than mature practice was now suggesting characterized SSM use, at least by those who had internalized it". It is the *Four Main Activities* version that is used to show 'the learning cycle of soft systems methodology' in *Information, Systems and Information Systems* (Checkland and Holwell, 1998b). In Checkland and Poulter's (2006) 'short definitive account of soft systems methodology and its use', the *Four Main Activities* version is described as having emerged as the 'classic representation' and as offering 'the iconic representation of SSM's learning cycle'. This 2006 version is pictured in Figure 5.

Figure 5. The iconic representation of SSM's learning cycle (Checkland and Poulter, 2006, p. 13)



We will now describe the four activities shown in Figure 5. In doing so, we must be aware of Checkland and Poulter's insistence that we are considering a 'learning cycle' that is never ending, and that work will likely be happening on all four activities simultaneously. This account follows that provided by Checkland and Poulter (2006).

The first activity involves *finding out* about the situation deemed as problematical. It is supported by making 'rich pictures' and by carrying out Analyses One, Two and Three. The second consists of making some *purposeful activity models*. The models should be relevant to the situation and are 'intellectual devices' each of which is constructed to capture a particular *Weltanshauung*. They are built with the help of 'root definitions' which have been shaped according to the PQR formula enriched by the elements captured in the CATWOE mnemonic. The models should include criteria for efficacy, efficiency, and effectiveness. The third activity uses the models to set up a comparison with what is perceived to exist in the problematical situation. Various ways of making this comparison are suggested. The aim is to structure a *discussion/debate* about possible improvements to the problematical situation. If an accommodation - an agreement that people can live with for the time being - can be reached on changes that are desirable and feasible, then *action to improve* the situation, the fourth activity, can be defined and implemented. There is another action, of course, at a meta-level to the other four – critical reflection on the process and the thinking employed based on the FMA framework.

At first sight, it might look like the 'stream of cultural analysis', given prominence in the *Two Streams* model, has been lost. This is not the case. As Checkland (1999, p. A15) explains, it has simply been subsumed in the four activities, where it is implied rather than declared.

Further contributions to the M, in the 2006 book, include reflections on the 'mind-set' needed in approaching an SSM study, an outline of some common misunderstandings of the nature of SSM, discussion of some craft skills that can ease its use, some further tips on using the techniques associated with the methodology, and answers to some 'frequently asked questions' about SSM. Checkland and Poulter want

to help people get out and try the M. In their view, group learning is best facilitated if an SSM study is conducted by those in the problematical situation, although facilitators with a sound knowledge of the approach can often be helpful.

Just prior to the publication of *Information, Systems and Information Systems,* Holwell (1997) had been doing some new thinking about the Constitutive Rules of SSM. She regarded the 1990 set as both 'too loose and not extensive enough'. They seemed to allow someone just using rich pictures to be doing SSM. At the same time, they were silent about some of its basic assumptions Her new attempt to answer the question 'what is SSM' is framed at three levels – the 'taken-as-given assumptions'; the 'process of inquiry'; and the 'elements' used in the process. The most significant statement is contained in the first of three 'taken-as-given-assumptions':

> (1) you must accept and act according to the assumption that social reality is socially constructed, continuously (Holwell, quoted in Checkland 1999, p. A35).

SSM could hardly be more firmly tied to the phenomenological and hermeneutic perspective, to what Burrell and Morgan called the 'interpretive paradigm'. Apparently, there can be no more learning in this respect. Checkland endorses Holwell's new Constitutive Rules in his 'Thirty Year Retrospective', saying that "this work gives us a solid basis for definitive constitutive rules for SSM" (Checkland, 1999, p. A35).

Checkland and Poulter, in *Learning for Action* (2006, p. 171), state that "more than likely … understanding the theory [underpinning SSM] will have a positive effect in improving any practitioner's use of the methodology". Now there is absolute certainty about that theory, Checkland and Holwell (1998b) and Checkland and Poulter (2006) can make exact statements about the F that sustains SSM. In a section on 'a concept of organization for IS', Checkland and Holwell reason that

> … an organization is clearly an abstraction … But what causes it, as an entity, to exist? The answer can only be: the *readiness* of some people, usually large numbers of people, members and non-members alike, to talk and act *as if* there were a collective entity

> which could behave like a conscious being, with the ability to decide to do things and then make them happen (p. 80; italics in the original).

In the same book, they state:

> … it was necessary to assume that social reality in human groups is continuously socially created in never-ending social processes, and hence is not an absolute but will change through time, sometimes slowly, sometimes very quickly (Checkland and Holwell 1998b, p. 157).

Checkland and Poulter are clear that:

> This shift from goal seeking/optimizing to sustaining relationships/learning turned out to be a much bigger shift than was at first realized. It is, in the language of academic social theory, a shift from one philosophy and sociology to a different philosophy and a different sociology. It is a move from positivism and functionalism (the 'hard' approach) to phenomenology and interpretive sociology (the 'soft' approach) (2006, p. 173).

As this quote suggests, it was the thinking of Vickers on 'relationship maintaining' that primarily inspired the shift. And it is his work on 'appreciative systems', Checkland says, that maps most completely experiences of using SSM:

> In summary, SSM can be seen as a systemic learning process which articulates the working of 'appreciative systems' in Vickers' sense (1999, p. A41).

In both Checkland 1999 (pp. A 50 – 58) and Checkland 2011(pp. 507-510) there are significant expositions of Vickers's 'appreciative systems theory' and its relationship to SSM.

Checkland has always emphasised that SSM must be used flexibly by the researcher according to the requirements of the problematical situation confronted. This led some, encouraged by Checkland, to argue that it could be employed in the service of a functionalist or radical change agenda if the researcher was so inclined. Given the clear commitment to the phenomenological and hermeneutic perspective, in later renditions of the Constitutive Rules, this option is emphatically closed. The researcher who does not accept the "assumption that social reality is socially constructed, continuously", and does not act accordingly, is not using SSM as conceived of by Checkland. Of course, a functionalist or radical change researcher can ignore Checkland and continue to use the methodology regardless. But, given

the way SSM has been rigorously developed to fit the view of reality championed by the interpretive paradigm, they will find that they are using a blunt instrument in pursuit of their favoured purposes. Better employ an approach specifically honed to those purposes. This is, I think, an important issue which, unfortunately, cannot be discussed further here.

With the establishment of the *Four Main Activities* model, Checkland regards SSM as 'mature'. This enables and justifies writing 'a short definitive account' of SSM (Checkland and Poulter 2006, p. 16) and qualifies the methodology to be included in the book *Systems Approaches to Making Change: A Practical Guide* (Checkland and Poulter 2020, p. 205). Evidence for its maturity, Checkland (2011 p.507) argues, is "a gradual diminution of activity" producing more learning about SSM following its use in problematical situations. This does not mean that there will be no more modifications based on experience, but it does indicate a slowdown in the rate of change.

The only recent addition to the literature on the M, that Checkland and Poulter feel it necessary to mention, comes from a 2006 paper by Checkland and Winter distinguishing its use to think about the *process* of carrying out a study (SSM 'p') and its use to explore the *content* of the perceived situation (SSM 'c'). And this, they say, is an example of the literature lagging behind practice.

In terms of the F, Checkland and Holwell (2004) develop the argument, first seen in Checkland (1985a), that the 'soft' and 'hard' systems approaches are 'asymmetrically complementary'. SSM is seen as relevant to all problematical situations while hard approaches are appropriate only in 'special cases', identified by SSM, when participants agree that it is desirable to engineer some system to achieve a goal as efficiently as possible.

A Review of the Development of Soft Systems Methodology (SSM) seen through the Lens of the FMA Framework

The development of SSM has been outlined using the FMA framework. The AR programme endorsed by Checkland is meant to yield improvements from the point of view of participants in the problem situation and enhancements to the methodology and ideas employed to make them more fit for purpose. It is now possible to review these matters in terms of success in *areas of application* (As) and changes to the *methodology* (M), and *framework of ideas* (F). As indicated in the introduction, the focus will be on the under-researched area of the development, or otherwise, of the F.

a. Areas of Application (A)

In a Postscript to Checkland and Poulter (2020, pp. 250-253), Checkland and colleagues can look back at over "300 well documented [SSM] interventions … undertaken to help bring about improvements in the real world and enable lessons to be learned". SSM has been readily accepted, by both academics and practitioners, as a successful approach to the management of unstructured problem situations and has spread its influence to many countries outside the UK. A 1992 paper by Mingers and Taylor demonstrated widespread use of SSM among OR and systems practitioners, with over 90% reporting their success with it as reasonable, good or very good. Ledington and Donaldson (1997) chart how widely it has been used in Australia, and with a similar success rating. Numerous case study papers demonstrate impact in a variety of fields – strategy development, health, information systems, knowledge management, education, agriculture, project management, performance management, and sustainable development. A scoping review of the use of SSM, and elements of SSM, in healthcare (Augustsson et al., 2020) categorizes 49 studies into six different types, including problem structuring, proposing improvements, and evaluation. A systematic literature review focussing on the use of SSM in the Information Systems field (Ali and Deneke, 2020) analysed 20 publications and found four main areas of application – organizational systems, education, E-service, and security functionalities. The experiences reported in the literature suggest that improvement to the A is a frequent outcome of employing SSM.

b. Methodology (M)

Any reader of the previous section cannot, I feel, be in any doubt that the methodology has moved on because of being employed in an AR mode. The shift from *Systems Engineering* to the *Blocks and Arrows* version was the most drastic but significant refinements continued to be made over the following decades. In Checkland's books and articles these are clearly linked to the learning gained from the projects undertaken.

As Checkland (1999, p. A11) sees it, agreeing with Holwell, the methodology became 'less structured and broader' as it developed. SSM gradually became a mature methodology capable of intervening in unstructured problematical situations when answering questions about 'what' should be done is of crucial significance. The clarity with which SSM was expressed at each stage of its development enabled the reflection that led to continual adaptation of the whole methodology as well as the parts, the methods and techniques, designed to support it. Checkland insists that

> … whether the changes to the parts were additions or deletions, they were never made by sitting at desks being 'academic'. They have always been made as a result of experiences in using the approach in a complex world … (Checkland 1999, p. A16).

While we can accept this in relation to the parts of SSM, it is less clear that it is true in relation to the 'whole', as we shall now explore.

c. Framework of Ideas (F)

The development of SSM's F is marked by an early period of exceptional change, when SE proved inadequate for more complex, management problems, followed by a long period of stability.

In the early days, Checkland was successful in taking Systems Thinking into a new paradigm and, in doing so, extended its scope to a much wider range of problem situations. SSM brings within the scope of systemic treatment all those 'wicked', messy, and ill-structured problem situations that either escape or are distorted by methodologies rooted in functionalism because of the strict prerequisites that need to

be met before those can be employed. Such a significant shift in the M was made possible, Checkland argues, because of the drastic failure of SE in complex problem situations:

> Even professional researchers who ought to be ready to welcome change in taken-as-given structures of thinking, show the same tendency to distort perceptions of the world rather than change the mental structures we use to give us our bearings. So we were lucky in our research programme that the failure of classic systems engineering in rich 'management' problem situations, broadly defined, was dramatic enough to send us scurrying to examine the adequacy of the systems thinking upon which systems engineering was based (Checkland 1999, p. A11).

Checkland, as we saw, is scrupulous in detailing the project experiences that contributed to this shift. It seems reasonable to suggest however, that it could not have been achieved so thoroughly if he had not, at the same time, been absorbing a range of new theoretical ideas. In accomplishing the break from the functionalist social theory implicit in SE, Checkland drew upon the work of Churchman, particularly his concept of *Weltanschauung,* Vickers's work on 'appreciative systems' and, a little later, Weber's sociology, Husserl's phenomenology, Dilthey's hermeneutics, and Burrell and Morgan's grid of sociological paradigms. It is the engagement with this work that enabled him to think through and bring theoretical clarity to his break with 'hard' systems thinking. His later writings do not, as a result, betray the same 'tensions' between the hard and soft positions that he identifies in socio-technical systems thinking and, for example, in the writings of Churchman and Ackoff. Checkland is the most theoretically pure of the soft systems thinkers because he recognised the theoretical direction the approach was taking, made this explicit, and began to consciously frame SSM employing alternative philosophical and sociological foundations.

SSM then became firmly rooted in interpretive sociology and this commitment only strengthened with time. It seems that Checkland was unable to challenge his 'mental structures' any further. The justification for this, he avows (1999, p. A40), is that the results of decades of AR using SSM provided confirmation of interpretive theory. The social world revealed itself to be as Husserl and Schutz, and especially

Vickers, describe it. Checkland insists that if, during his experiences using SSM, he had found other theoretical perspectives useful, he would have learned from them. The trouble is that, as Checkland noted, taken-for-granted mental structures easily become fixed whatever the evidence. Or, as Churchman would put it, no data can ever destroy a *Weltanschauung*. Unless you are willing and able to view the world through different theoretical lenses (as Checkland was at the time of abandoning SE), you are likely to go on finding confirmation for the one you favour, however much real world experience you gain. Checkland does not seek to make use of alternative theoretical perspectives. Indeed, following *Systems Thinking, Systems Practice* (1981), he shows little interest in exploring at any depth what different forms of social theory have to offer.

Checkland's embrace of interpretive sociology as a fixed F for SSM helped to produce a refined and useful M entirely suited to that view of the social world. There have, however, been consequences. Two of the possibilities held out for SSM, in 1991, failed to materialize – the discovery of "common structurings which characterize the social collectivities within which it works" and its possible use as an instrument of radical change.

It was left to Brian Wilson, Checkland's collaborator at Lancaster, operating with a version of SSM that Checkland (1999, p.A4) describes as cleaving "to the functional logic of engineering", to unearth various models of social processes which, he believes, have wide applicability. In his early work, Wilson (1984) tried to follow the party line on SSM. However, in a more recent book, with Van Haperen (2015), fundamental differences with Checkland have become clear. Wilson and Van Haperen argue that all organizations must have a core transformation system, supporting systems, linking systems, and adaptive elements. These will share core characteristics in many organizations. It is therefore possible to produce generic models that are useful in a range of settings. Thus, they present generic models for ISO 9001, Programme Realisation, Vision Realisation, Logistics, Change Management, Benefits Management, A Generic Company

Model, and 'An Issue-based model relevant to respond to an Information-To-Tender' (Wilson and Van Haperen 2015, p. 84). This is in fulfilment of Jenkins's original hope that the AR programme would lead to generic models exactly of this kind. Wilson and Van Haperen (2015, p. lxxvii) accept that their approach shows a 'proximity' to SE but argue that in many instances in the real world this is exactly what is required. In practical application, in areas such as organizational design and change, procurement, information systems, and engineering, they find the need for more elaborate models that can be used for the purpose of design. It seems that these two successful consultants have been led, by their version of SSM, to a rather different understanding of the nature of social reality than Checkland, who can find no evidence of 'common structurings' that could provide a solid basis for functionalist approaches. Wilson and Van Haperen even suggest that a form of groupthink in the academic community, encouraged by the proponents and gatekeepers of Checkland's version of SSM, has played a part in their experiences being ignored and their voices silenced.

The claim that SSM could ever be a vehicle for bringing about radical change has been much criticised. The critics argue that it lacks a social theory capable of considering the impact on its usefulness of structural inequalities and prevailing power structures. As Willmott has it:

> Its major shortcoming lies in its unnecessarily limited capacity to promote reflection upon the possibility that the content and negotiation of *Weltanschauungen* are expressive of asymmetrical relations of power through which they are constructed and debated … Phenomenology, and SSM in particular, simply lacks a social theory capable of accounting for why particular sets of perceptions of reality emerge, and why some perceptions are found more plausible than others (Wilmott 1989, p.76).

Mingers (1984) argues that it is the subjectivism inherent in interpretive sociology that stands in the way of SSM developing the necessary social theory. It is condemned by this to act at the level of ideas, seeking to change things by changing worldviews. It is ignorant of how particular social arrangements and power structures give rise to

worldviews and determine their influence. From a radical change perspective

> …it could be said that SSM, in focusing exclusively on the espoused beliefs and values of individual people, thereby lost connection to the wider social and political structure that shaped such beliefs (Mingers 2000, p. 743).

There is, of course, some recognition of power in the *Two Streams* version of SSM. This, however, is only in terms of understanding the 'commodities' of power available to individuals in a particular organization, not as something more pervasive and inextricably entwined with structures of inequality in the broader social system. Checkland wants an account of power that can be used for immediate practical purposes and is compatible with the interpretive nature of SSM. As Mingers puts it:

> In Chapter 2 [of *Soft Systems Methodology in Action*] the need for social and political analysis within the methodology is recognized … the response is not to look in the general literature of social and political theory, but to pick up on quite marginal work by Vickers … on 'appreciative systems' and a Lancaster Ph.D. thesis by Stowell … on 'commodities of power' (Mingers 2000, p. 747-8).

The impact of this on what can be achieved using SSM has become apparent. Clarke and Lehaney (1999) and Connell (2001) encountered significant problems using SSM in the face of prevailing power structures. They were able to understand why by looking at the world through the lens of radical change theory. Checkland's commitment to interpretive sociology means that he cannot see the value of any M based on different theoretical foundations, except as strictly subordinated to his own approach (as in stage 4b of the *Seven-Stages* version). In practice, Mingers (2000, p. 748) cannot find "a single example from Checkland's writing where any methodology other than SSM has been used – and this is a great lost opportunity". Another consequence is that Checkland cannot reflect on just how 'regulative' many of his interventions look from alternative theoretical perspectives. For example, two of his best known cases, with the 'manufacturing function' of the Shell Group (Checkland and Scholes 1990; Checkland and Poulter 2006) and with an Information Department in the central research and development laboratories of a

multi-national science-based group (Checkland and Holwell 1998b), are easily read as simply regulating support departments so that they function more appropriately in terms of the purpose of the overall system.

Conclusions

In conclusion, it is worth considering what lessons can be learned from our analysis of SSM, using the FMA framework, for other systems methodologies and for applied systems thinking generally.

The relationship between developments in SSM's M and the increasingly diverse As in which it was employed represents an exemplar of how the FMA framework should work. Increasing experience in using the M led to a range of modifications which equipped the M to be more effective in a wider range of problematical situations. As we have seen, there were significant changes to the M - drastic in the early days, more gradual later – over decades of use in over 300 AR projects. The 'recoverability' of these changes from the relevant literature, and the fact that they can be directly linked to the experience of application, have led to understanding and full acceptance of the 'mature' version of SSM within the relevant academic and practitioner communities.

The argument becomes a bit more complex when we come to consider the role of the F in the FMA framework. Our analysis has shown that experience of using SE in complex social situations produced learning that led to a radical change in the M and a paradigm shift in the underlying F. The rejection of functionalist thinking, and the embrace of phenomenology and hermeneutics, supported the further development of the M and widened the range of As in which systems thinking could be successfully employed. However, from that point, and certainly since SSM became 'mature', there has been little further development, or even questioning, of the F. As Checkland, and some close colleagues, noted:

> Nowadays, it seems that SSM's widespread adoption by practitioners around the world, as evidenced through its publications, proceeds more rapidly than new academic development of its theoretical foundations and principles (Checkland 2020, p. 252).

The question is whether this is a good or bad thing.

There are great benefits to be had from research communities sticking to a preferred F and pursuing 'normal science', as Kuhn (1970) calls it, using their traditional M. The full details of the theory can be worked out and the possibilities it offers fully explored. The M can be refined to better serve the F in translating its insights into practice. Methods and techniques can be devised and perfected to support an M in use. This is certainly what happened in the case of SSM. It can be seen as a strength of SSM, therefore, that it developed and stuck to a powerful and coherent F. It has provided it with a unique competence. It can and should, within the phenomenological and hermeneutic tradition, continue to develop its M and its proficiency in As. This is what I read Stowell (2021a, 2021b, 2024) as seeking to achieve.

I have argued elsewhere (Jackson 2025) that Checkland's FMA framework provides a useful vehicle for understanding the development of a range of applied systems approaches – systems engineering, system dynamics, the viable system model, critical systems heuristics, as well as SSM. And that following its dictates is the best way for advocates of those approaches to continue to enhance their own theories, methodologies, models, methods and techniques and, as a result, improve their practical effectiveness. As our exposition of the development of SSM has hopefully shown, adopting the FMA framework is an excellent way of doing so.

On the other hand, critics might argue, with Churchman, that, unless you can look at the world through alternative *Weltanshauungen*, you will go on finding confirmation for the one to which you have come to subscribe. They will argue that Checkland, once committed to interpretive social theory, could no longer see what functionalists could plainly see and could not respond to what radical change thinkers regarded as important. However, this only becomes an issue if you expect SSM to do everything, to offer a response to all forms of

complexity. But it cannot do this and, if it tried, would become a theoretical and practical mess and lose its unique competence. When SSM established its own F, M and As, it enhanced systems thinking. Nothing was lost because SE remained strong and continued to be successful in its own domain of capability. Similarly, system dynamics, the viable system model, and critical systems heuristics, have their own domains of competence. There is great value in each of these systems communities continuing to develop and refine their own theories and competences.

For the sake of applied systems thinking generally, however, it is essential that each of those communities come to accept that their favoured systems approach only offers a partial view of the world and a limited way of intervening in problem situations. Systems Thinking as a field will only reach its full potential when it is prepared to entertain a variety of Fs and Ms and, thereby, makes itself relevant to the widest possible range of As. Systems methodologies must be used in tandem to address multi-dimensional complexity – an approach to systems practice championed by Critical Systems Thinking (Jackson 2019, 2024, 2025).

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